

VISVESVARAYA TECHNOLOGICAL UNIVERSITY



BELAGAVI – 590018, Karnataka

INTERNSHIP REPORT ON

“Chatbot for Healthcare system using AI”

Submitted in partial fulfilment for the award of degree (21CSI85)

BACHELOR OF ENGINEERING IN COMPUTER SCIENCE & ENGINEERING

Submitted by:

ANUSHA R D

1EE22CS400



Conducted at
COMPSOFT TECHNOLOGIES



EAST WEST COLLEGE OF ENGINEERING
Department of Computer Science & Engineering
Accredited by VTU, BELAGAVI

EAST WEST COLLEGE OF ENGINEERING
Department of Computer Science & Engineering
Accredited by VTU, Belagavi



CERTIFICATE

This is to certify that the Internship titled “**Chatbot for Health system using AI**” carried out by **MS. ANUSHA R D**, a bonafide student of East West College of engineering, in partial fulfillment for the award of **Bachelor of Engineering, in Computer Science & Engineering** under Visvesvaraya Technological University, Belagavi, during the year 2023-2024. It is certified that all corrections/suggestions indicated have been incorporated in the report.

The project report has been approved as it satisfies the academic requirements in respect of Internship prescribed for the course Internship / Professional Practice (21CSI85)

Signature of Guide

Signature of HOD

Signature of Principal

External Viva:

Name of the Examiner

Signature with Date

1) _____

2) _____

D E C L A R A T I O N

I, **ANUSHA R D**, III year student of Computer Science and Engineering, East West College of Engineering, declare that the Internship has been successfully completed, in **COMPSOFT TECHONOLOGIES**. This report is submitted in partial fulfillment of the requirements for award of Bachelor Degree in Branch name, during the academic year 2023-2024.

Date:

Place: Bangalore

USN: 1EE22CS400

NAME: ANUSHA R D

OFFER LETTER



Internship Offer Letter

Date: 25th October, 2023

Name: **ANUSHA R D**

USN: **1EE22CS400**

Placement ID: **23OCTMLBONE**

Dear Student,

We would like to congratulate you on being selected for the **Machine Learning with Python (Research Based)** Internship position with **Compsoft Technologies**, effective Start Date **25th October, 2023**, All of us are excited about this opportunity provided to you!

This internship is viewed as being an educational opportunity for you, rather than a part-time job. As such, your internship will include training/orientation and focus primarily on learning and developing new skills and gaining a deeper understanding of concepts of **Machine Learning with Python (Research Based)** through hands-on application of the knowledge you learn while you train with the senior developers. You will be bound to follow the rules and regulations of the company during your internship duration.

Again, congratulations and we look forward to working with you!.

Sincerely,

Nitin K. S

Project Manager

Compsoft Technologies

No. 363, 19th main road,

1st Block Rajajinagar Bangalore -

560010

A C K N O W L E D G E M E N T

This Internship is a result of accumulated guidance, direction and support of several important persons. We take this opportunity to express our gratitude to all who have helped us to complete the Internship.

We express our sincere thanks to our Principal, for providing us adequate facilities to undertake this Internship.

We would like to thank our Head of Dept. – branch code, for providing us an opportunity to carry out Internship and for his valuable guidance and support.

We would like to thank our (Lab assistant name) Software Services for guiding us during the period of internship.

We express our deep and profound gratitude to our guide, Guide name, Assistant/Associate Prof, for her keen interest and encouragement at every step in completing the Internship.

We would like to thank all the faculty members of our department for the support extended during the course of Internship.

We would like to thank the non-teaching members of our dept., for helping us during the Internship.

Last but not the least, we would like to thank our parents and friends without whose constant help, the completion of Internship would have not been possible.

ANUSHA R D
1EE22CS400

ABSTRACT

As the demand in Machine Learning and AI keeps growing, new technologies will keep coming in the market which will impact our day-to-day activities, and one such technology is Virtual Assistant Bots or simply Chatbots. Chatbots have evolved from being Menu/Button based, to Keywords based and now Contextual based. The most advanced among all of the above is contextual based because it uses Machine Learning and Artificial Intelligence techniques to store and process the training models which help the chatbot to give better and appropriate response when user asks domain specific questions to the bot.

A chatbot is an artificial intelligence computer program which performs communication using text, audio and video systems. A person can ask any questions and chatbot will answer accordingly. The main objective of this research work is to create a web UI HealthCare Chatbot, and also about sample web and text messaging interfaces that demonstrate the use of API.

Healthcare is very important to lead a good life. However, it is very difficult to obtain the consultation with the doctor for every health problem. The idea is to create a medical chatbot using Artificial Intelligence that can diagnose the disease and provide basic details about the disease before consulting a doctor. This will help to reduce healthcare costs and improve accessibility to medical knowledge through medical chatbot. The chatbots are computer programs that use natural language to interact with users. The chatbot stores the data in the database to identify the sentence keywords and to make a query decision and answer the question. Ranking and sentence similarity calculation is performed using n-gram, TF-IDF and cosine similarity. The score will be obtained for each sentence from the given input sentence and more similar sentences will be obtained for the query given. The third party, the expert program, handles the question presented to the bot that is not understood or is not present in the database

The idea is to create a medical chatbot that can diagnose the disease and provide basic details about the disease before consulting a doctor. This will help to reduce healthcare costs and improve accessibility to medical knowledge through medical chatbot. The chatbots are computer programs that use natural language to interact with users. Our project focuses on providing the users immediate and accurate prediction of the diseases based on their symptoms. For the prediction of diseases, we have used Decision tree algorithm. Chatbots can play a major role in reshaping the healthcare industry by providing predictive diagnosis.

Table of Contents

Sl no	Description	Page no
1	Company Profile	8
2	About the Company	10
3	Introduction	14
4	Literature Survey	17
5	System Analysis	19
6	Requirement Analysis	21
7	Design Analysis	23
8	Implementation	25
9	Snapshots	37
10	Conclusion	45
11	References	47

CHAPTER 1

COMPANY PROFILE

1. COMPANY PROFILE

A Brief History of Compsoft Technologies

Compsoft Technologies, was incorporated with a goal” To provide high quality and optimal Technological Solutions to business requirements of our clients”. Every business is a different and has a unique business model and so are the technological requirements. They understand this and hence the solutions provided to these requirements are different as well. They focus on clients requirements and provide them with tailor made technological solutions. They also understand that Reach of their Product to its targeted market or the automation of the existing process into e-client and simple process are the key features that our clients desire from Technological Solution they are looking for and these are the features that we focus on while designing the solutions for their clients.

Sarvamoola Software Services. is a Technology Organization providing solutions for all web design and development, MYSQL, PYTHON Programming, HTML, CSS, ASP.NET and LINQ. Meeting the ever increasing automation requirements, Sarvamoola Software Services. Specialize in ERP, Connectivity, SEO Services, and Conference Management, effective web promotion and tailor-made software products, designing solutions best suiting client’s requirements.

Compsoft Technologies, strive to be the front runner in creativity and innovation in software development through their well-researched expertise and establish it as an out of the box software development company in Bangalore, India. As a software development company, they translate this software development expertise into value for their customers through their professional solutions.

They understand that the best desired output can be achieved only by understanding the clients demand better. Compsoft Technologies work with their clients and help them to define their exact solution requirement. Sometimes even they wonder that they have completely redefined their solution or new application requirement during the brainstorming session, and here they position themselves as an IT solutions consulting group comprising of high caliber consultants.

They believe that Technology when used properly can help any business to scale and achieve new heights of success. It helps Improve its efficiency, profitability, reliability; to put it in one sentence ” Technology helps you to Delight your Customers” and that is what we want to achieve.

CHAPTER 2

ABOUT THE COMPANY

2. ABOUT THE COMPANY



Compsoft Technologies is a Technology Organization providing solutions for all web design and development, MYSQL, PYTHON Programming, HTML, CSS, ASP.NET and LINQ. Meeting the ever increasing automation requirements, Compsoft Technologies specialize in ERP, Connectivity, SEO Services, Conference Management, effective web promotion and tailor-made software products, designing solutions best suiting clients requirements. The organization where they have a right mix of professionals as a stakeholders to help us serve our clients with best of our capability and with at par industry standards. They have young, enthusiastic, passionate and creative Professionals to develop technological innovations in the field of Mobile technologies, Web applications as well as Business and Enterprise solution. Motto of our organization is to “Collaborate with our clients to provide them with best Technological solution hence creating Good Present and Better Future for our client which will bring a cascading a positive effect in their business shape as well”. Providing a Complete suite of technical solutions is not just our tag line, it is Our Vision for Our Clients and for Us, We strive hard to achieve it.

Products of Compsoft Technologies.

Android Apps

It is the process by which new applications are created for devices running the Android operating system. Applications are usually developed in Java (and/or Kotlin; or other such option) programming language using the Android software development kit (SDK), but other development environments are also available, some such as Kotlin support the exact same Android APIs (and bytecode), while others such as Go have restricted API access.

The Android software development kit includes a comprehensive set of development tools. These include a debugger, libraries, a handset emulator based on QEMU, documentation, sample code, and tutorials. Currently supported development platforms include computers running Linux (any modern desktop Linux distribution), Mac OS X 10.5.8 or later, and Windows 7 or later. As of March 2015, the SDK is not available on Android itself, but software development is possible by using specialized Android applications.

Web Application

It is a client–server computer program in which the client (including the user interface and client- side logic) runs in a web browser. Common web applications include web mail, online

retail sales, online auctions, wikis, instant messaging services and many other functions. web applications use web documents written in a standard format such as HTML and JavaScript, which are supported by a variety of web browsers. Web applications can be considered as a specific variant of client–server software where the client software is downloaded to the client machine when visiting the relevant web page, using standard procedures such as HTTP. The Client web software updates may happen each time the web page is visited. During the session, the web browser interprets and displays the pages, and acts as the universal client for any web application. The use of web application frameworks can often reduce the number of errors in a program, both by making the code simpler, and by allowing one team to concentrate on the framework while another focuses on a specified usecase. In applications which are exposed to constant hacking attempts on the Internet, security-related problems can be caused by errors in the program.

Frameworks can also promote the use of best practices such as GET after POST. There are some who view a web application as a two-tier architecture. This can be a “smart” client that performs all the work and queries a “dumb” server, or a “dumb” client that relies on a “smart” server. The client would handle the presentation tier, the server would have the database (storage tier), and the business logic (application tier) would be on one of them or on both. While this increases the scalability of the applications and separates the display and the database, it still doesn’t allow for true specialization of layers, so most applications will outgrow this model. An emerging strategy for application software companies is to provide web access to software previously distributed as local applications. Depending on the type of application, it may require the development of an entirely different browser-based interface, or merely adapting an existing application to use different presentation technology. These programs allow the user to pay a monthly or yearly fee for use of a software application without having to install it on a local hard drive. A company which follows this strategy is known as an application service provider (ASP), and ASPs are currently receiving much attention in the software industry.

Security breaches on these kinds of applications are a major concern because it can involve both enterprise information and private customer data. Protecting these assets is an important part of any web application and there are some key operational areas that must be included in the development process. This includes processes for authentication, authorization, asset handling, input, and logging and auditing. Building security into the applications from the beginning can be more effective and less disruptive in the long run.

Web design

It encompasses many different skills and disciplines in the production and maintenance of websites. The different areas of web design include web graphic design; interface design; authoring, including standardized code and proprietary software; user experience design; and

search engine optimization. The term web design is normally used to describe the design process relating to the front-end (client side) design of a website including writing mark up. Web design partially overlaps web engineering in the broader scope of web development. Web designers are expected to have an awareness of usability and if their role involves creating mark up then they are also expected to be up to date with web accessibility guidelines. Web design partially overlaps web engineering in the broader scope of web development.

Departments and services offered

Compsoft Technologies plays an essential role as an institute, the level of education, development of student's skills are based on their trainers. If you do not have a good mentor then you may lag in many things from others and that is why we at Compsoft Technologies gives you the facility of skilled employees so that you do not feel unsecured about the academics. Personality development and academic status are some of those things which lie on mentor's hands. If you are trained well then you can do well in your future and knowing its importance of Compsoft Technologies always tries to give you the best.

They have a great team of skilled mentors who are always ready to direct their trainees in the best possible way they can and to ensure the skills of mentors we held many skill development programs as well so that each and every mentor can develop their own skills with the demands of the companies so that they can prepare a complete packaged trainee.

Services provided by Compsoft Technologies.

- Core Java and Advanced Java
- Web services and development
- Dot Net Framework
- Python
- Selenium Testing
- Conference / Event Management Service
- Academic Project Guidance
- On The Job Training
- Software Training

CHAPTER 3

INTRODUCTION

3. INTRODUCTION

Introduction to ML

Arthur Samuel, an early American leader in the field of computer gaming and artificial intelligence, coined the term “Machine Learning” in 1959 while at IBM. He defined machine learning as “the field of study that gives computers the ability to learn without being explicitly programmed “. However, there is no universally accepted definition for machine learning. Different authors define the term differently. We give below two more definitions.

Machine learning is programming computers to optimize a performance criterion using example data or past experience. We have a model defined up to some parameters, and learning is the execution of a computer program to optimize the parameters of the model using the training data or past experience. The model may be predictive to make predictions in the future, or descriptive to gain knowledge from data. The field of study known as machine learning is concerned with the question of how to construct computer programs that automatically improve with experience.

Machine learning is a subfield of artificial intelligence that involves the development of algorithms and statistical models that enable computers to improve their performance in tasks through experience. These algorithms and models are designed to learn from data and make predictions or decisions without explicit instructions. There are several types of machine learning, including supervised learning, unsupervised learning, and reinforcement learning. Supervised learning involves training a model on labeled data, while unsupervised learning involves training a model on unlabeled data. Reinforcement learning involves training a model through trial and error. Machine learning is used in a wide variety of applications, including image and speech recognition, natural language processing, and recommender systems.

Problem Statement

At the start of each academic semester, registration opens for those wishing to join the university in various disciplines, and telephone calls for admission and registration abound. This leads to an increase in the loads and work for the employees of the Deanship of Admission and Registration as a result of the constant pressure of those wishing to register and their families by flocking to the Deanship, so the employees are not able to answer the phone calls and social media. This often leads to many students who wish to register to be ignored.

Making a chatbot ready to respond to each and every inquiry is absurd to expect to execute with current innovation and inside the span of the task, so the framework will have the option to address inquiries regarding restricted themes.

Language: The framework will just help inquiries in Standard English.

Datasets: Giving support for wide varieties of Topics will require writing more modules, which is quite challenging for doing Individually.

For a long time, researchers in artificial intelligence (AI) have been intrigued by the idea of developing a conversational agent that is capable of having a coherent conversation with humans. Recent breakthroughs in semantics and speech recognition have given rise to hopes for robust solutions to the problem. Major information technology companies have released digital assistants and chatbot frameworks to facilitate the building of conversational agents.

However, research studies on the user perception and expectations from users of chatbots indicate that the systems still require significant improvements in order to provide a meaningful experience. Also, demand analysis identified a need for specialized digital assistants in customer facing processes, in particular in the insurance sector. A promising approach to advance in this field is comprehensive knowledge engineering methodologies which back the chatbot and upgrade its conversational capabilities from small talk to domain expert.

Ever since the COVID-19 pandemic started there has been a drastic change in everyone's life. The pandemic has paved the way for new problems to arise. Among them, one is to visit the doctor's office without getting infected. Many times, many people fear going to the doctor for a normal checkup because of COVID-19. The coronavirus pandemic has made us use the virtual space more. WFH has been implemented, home delivery is increasing and the reasons that people have for going out has become considerably low. So why not create a robot which can help you diagnose your normal health related issues without physically going to hospital.

Chat has become the center of focus in this current era; thus, the bots are being utilized to deliver information engagingly and conveniently. A chatbot is standout amongst the most progressive and promising tools of communication among people and machines. Famous chatbots like Google Assistant, Amazon Alexa, Siri, Facebook, Slack, and many more are in trend. These are very helpful, but in this era of enhancing technology, day by day technology gets updated, and accordingly, user expectations also increase. A user wants more automation in the chatbot. Although every system is not perfect there is always a flaw in the system, so as in the chatbot there are some problems that the user has experienced while using a chatbot.

A bot is trained on and according to the training, based on some rules on which it is trained, it answers questions. It is called a ruled based approach. Using these ruled based approach, creation of these bots becomes relatively straight forward. But it is not sufficient for the bot to answer questions whose pattern does not match the rules on which it is trained. The language by which these bots can be created is Artificial Intelligence Markup Language (AIML). It is a language based on XML which allows the developer to write the rules which bot will follow.

CHAPTER 4

LITERATURE SURVEY

4. Literature Survey

[1] An application of counseling chatbot, which provides conversational service for mental health care based on emotions recognition methods and chat assistant platform. This application doesn't consider the user's psychiatric status through continuous user monitoring.

[2] In this text-based healthcare chatbots can be designed to effectively support patients and health professionals in therapeutic settings beyond on side consultations. It does not have face to face care and where THCB are likely to fail.

[3] This paper reports a primary care chatbot system created to assist healthcare staff by automating the patient's intake process. In his paper they did not include more disease and also a symptom synonym thesauri also should be produced.

[4] This provides a text-to-text conversational age that asks the user about their health issue. The user can chat as if chatting with a human. The bot then asks the user a series of questions about their symptoms to diagnose and gives suggestions about the different symptoms to clarify the disease. Doesn't give detailed information No features such as duration, intensity of symptoms etc.

[5] The proposed method is a chatbot based mobile healthcare service that can immediately respond to the accidents that arise in everyday life and to the condition changes of chronic-disease patients. Also proposes a framework for the human-robot interaction that can endure an efficient implementation of the chatbot service. It is a text based bot irrespective of having all advanced features.

[6] Conversational agents have many technical, design and linguistic challenges. They introduced the nature of conversation user interface (CUI) for health and described UX design. Some technical limitations like voice message are not accurate, some corruptions are faced due to the network so the timing of bot remainders is corrupted.

[7] The Bot Transition program provides a framework and resources based on AAP, AFP and ACP recommendations to promote skill attainment in self-care. A scripted text messaging platform is feasible and appears to be well-received by patients and caregivers. It is designed only for people with special health needs transitioning into adolescents.

[8] The system uses a question and answer protocol in the form of a chatbot to answer users' queries. The complex questions and answers present in the database are viewed and answered by an expert. This chatbot is comparatively time consuming.

[9] This chatbot is an attempt to let users understand the symptoms they are facing and get a basic diagnosis about the disease they could be having. Complex interface, time consuming, high installation cost.

[10] A chatbot is an interactive software application to simulate natural user interactions based on AI modeling. We have proposed a dataset for the commonly occurring medical conditions together with a prototype model to provide quick assistance to the patients. It only gives solutions for the common medical conditions.

[11] Uses both KNN algorithm and decision tree classifier and from that more accurate one is taken and shows the output. As it uses both algorithms it is time consuming

CHAPTER 5

SYSTEM ANALYSIS

5. SYSTEM ANALYSIS

1. Existing System

Many of the existing systems have chats through texts. Some limitations of such Chatbots are, there is no instant response given to the patient, they have to wait for experts acknowledgement for a long time. And also there are a limited number of diseases in the dataset. Technical issues like voice messages are not accurate in the existing system.

2. Proposed System

In our system the user can interact with the chatbot through text and chat bot will interact using voice and text manner. With respect to the users queries, the bot identifies the disease if user chatting with the chatbot. According to the diseases of the user, bot gives suggestions for the disease and also prescribe specialist doctors. This system can be used by multiple users at a time without any lagging.

3. Objective of the System

- To extract symptoms from user chat.
- To classify and predict the diseases using a decision tree classifier.
- To develop a healthcare chatbot to predict diseases by symptoms taken as input.
- Save effort and time for both the Admission and registration staff and students who wish to enroll.
- Provide detailed information about colleges and majors.
- Easy access to information.

4. Scope

- People who wish to enroll Palestine Polytechnic University
- Admission and registration staff

With the chatbot, one doesn't have to wait to talk to the customer helpline, they don't even have to search for shopping through Websites. A chatbot is used in many areas like ordering food, product suggestions, customer support, weather, personal finance assistance, scheduled a meeting, search and track flights, send money, and many more. The main objective that we will discuss in this paper is creating a web UI HealthCare Chatbot, and also about sample web and text messaging interfaces that demonstrate the use of API. In this research paper we are trying to understand these Chatbots and understand their shortcomings. The technology at the core of the rise of the chatbot is natural language processing 'NLP'. Recent advances in machine learning have greatly improved accuracy and effectiveness natural language processing, making chatbots a viable option for many organizations. This improvement in NLP is firing a great deal of additional research which should lead to be improvement in the effectiveness of chatbots in the years to come. The datasets are taken online. Most commercial chatbots are dependent on platforms created by the technology giants for their natural language processing. These include Amazon Lex, Microsoft service Services, Google Cloud Natural Language API, Facebook Deep Text, and IBM Watson. Platforms where chatbots are deployed include Facebook Messenger, Skype, and Slack, among many others. The objective of this work is to create a chatbot that can answer your health-related problems. It has an effective User Interface and answers your health-related questions efficiently.

CHAPTER 6

REQUIREMENT ANALYSIS

6. REQUIREMENT ANALYSIS

The Hardware and Software Requirements are as follows.

Hardware Requirement Specification

- **Device name:** DESKTOP-46KDRGI
- **Processor :** Intel(R) Core(TM) i5-5300U CPU @ 2.30GHz 2.29 GHz
- **RAM :** 8.00 GB (7.70 GB usable)
- **Device ID :** DBD6D918-55AE-41C4-A926-EBE948CF57FF
- **Product ID :** 00330-80000-00000-AA832
- **System type :** 64-bit operating system, x64-based processor

Software Requirement Specification

- **Operating System :** windows XP
- **Programming language :** python
- **Ide :** vs code
- **AI tools :** Natural language processing
- **Sources :** GitHub

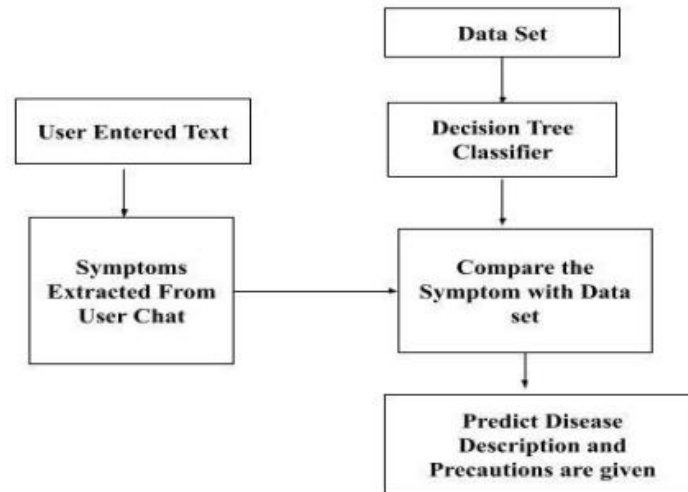
CHAPTER 7

DESIGN ANALYSIS

• DESIGN & ANALYSIS

SYSTEM DESIGN

The notion here is that the Naive Bayes classifier will predict the label based on the input we give it. So when you say 'hi' our classifier will predict the label '1', which in return we can use to find a suitable answer. When the input is 'what's your age?' classifier will predict the label '3', which is an index of the answer 'I'm 22 years old'. The below figure will dissolve any misgivings you have about the notion.



RESULT ANALYSIS

The proposed system is fully automated. The query or issue will be answered basis the question and knowledge base automatically. Hence there is no need to have a person (Doctor) to answer the queries and ease for the patients to interact with the queries. Patients can interact with the healthcare Chatbot in a very simple manner. The focus now is on developing the bot in a more intellectual way that it can even understand not so well grammatically defined sentences. Bot uses Chatterbot library, pattern matching and Naïve Bayers Technique to parse the patient's issues.

CHAPTER 8

IMPLEMENTATION

8.IMPLEMENTATION

Implementation is the stage where the theoretical design is turned into a working system. The most crucial stage in achieving a new successful system and in giving confidence on the new system for the users that it will work efficiently and effectively.

The system can be implemented only after thorough testing is done and if it is found to work according to the specification. It involves careful planning, investigation of the current system and its constraints on implementation, design of methods to achieve the change over and an evaluation of change over methods a part from planning.

Two major tasks of preparing the implementation are education and training of the users and testing of the system. The more complex the system being implemented, the more involved will be the system analysis and design effort required just for implementation.

The implementation phase comprises of several activities. The required hardware and software acquisition is carried out. The system may require some software to be developed. For this, programs are written and tested. The user then changes over to his new fully tested system and the old system is discontinued.

TESTING

The testing phase is an important part of software development. It is the Information zed system will help in automate process of finding errors and missing operations and also a complete verification to determine whether the objectives are met and the user requirements are satisfied. Software testing is carried out in three steps:

1. The first includes unit testing, where in each module is tested to provide its correctness, validity and also determine any missing operations and to verify whether the objectives have been met. Errors are noted down and corrected immediately.
2. Unit testing is the important and major part of the project. So errors are rectified easily in particular module and program clarity is increased. In this project entire system is divided into several modules and is developed individually. So unit testing is conducted to individual modules.
3. The second step includes Integration testing. It need not be the case, the software whose modules when run individually and showing perfect results, will also show perfect results when run as a whole.

Code

```
import random
from flask import jsonify
import secrets
from flask import Flask, render_template, flash, redirect, url_for, session, logging, request, session
from flask_sqlalchemy import SQLAlchemy

ALLOWED_EXTENSIONS = ['png', 'jpg', 'jpeg']
app = Flask(__name__)
app.config['SQLALCHEMY_DATABASE_URI'] = 'sqlite:///database.db'
db = SQLAlchemy(app)
app.secret_key = "m4xpl0it"
app.app_context().push()

def make_token():
    """
    Creates a cryptographically-secure, URL-safe string
    """
    return secrets.token_urlsafe(16)

class user(db.Model):
    id = db.Column(db.Integer, primary_key=True)
    username = db.Column(db.String(80))
    email = db.Column(db.String(120))
    password = db.Column(db.String(80))

@app.route("/")
def index():
    return render_template("index.html")

userSession = {}

@app.route("/user")
def index_auth():
    my_id = make_token()
    userSession[my_id] = -1
    return render_template("index_auth.html", sessionId=my_id)

@app.route("/instruct")
def instruct():
    return render_template("instructions.html")

@app.route("/upload")
def bmi():
    return render_template("bmi.html")
```

```
@app.route("/diseases")
def diseases():
    return render_template("diseases.html")

@app.route('/pred_page')
def pred_page():
    pred = session.get('pred_label', None)
    f_name = session.get('filename', None)
    return render_template('pred.html', pred=pred, f_name=f_name)

@app.route("/login", methods=["GET", "POST"])
def login():
    if request.method == "POST":
        uname = request.form["uname"]
        passw = request.form["passw"]

        login = user.query.filter_by(username=uname, password=passw).first()
        if login is not None:
            return redirect(url_for("index_auth"))
    return render_template("login.html")

@app.route("/register", methods=["GET", "POST"])
def register():
    if request.method == "POST":
        uname = request.form['uname']
        mail = request.form['mail']
        passw = request.form['passw']

        register = user(username=uname, email=mail, password=passw)
        db.session.add(register)
        db.session.commit()

        return redirect(url_for("login"))
    return render_template("register.html")

import msgConstant as msgCons
import re

all_result = {
    'name': "",
    'age': 0,
    'gender': "",
    'symptoms': []
}
```

```
# Import Dependencies
# import gradio as gr
import pandas as pd
import numpy as np
from joblib import load
import numpy as np
from sklearn.metrics.pairwise import cosine_similarity

def predict_symptom(user_input, symptom_list):
    # Convert user input to lowercase and split into tokens
    user_input_tokens = user_input.lower().replace("_", " ").split()

    # Calculate cosine similarity between user input and each symptom
    similarity_scores = []
    for symptom in symptom_list:
        # Convert symptom to lowercase and split into tokens
        symptom_tokens = symptom.lower().replace("_", " ").split()

        # Create count vectors for user input and symptom
        count_vector = np.zeros((2, len(set(user_input_tokens + symptom_tokens))))
        for i, token in enumerate(set(user_input_tokens + symptom_tokens)):
            count_vector[0][i] = user_input_tokens.count(token)
            count_vector[1][i] = symptom_tokens.count(token)

        # Calculate cosine similarity between count vectors
        similarity = cosine_similarity(count_vector)[0][1]
        similarity_scores.append(similarity)

    # Return symptom with highest similarity score
    max_score_index = np.argmax(similarity_scores)
    return symptom_list[max_score_index]
```

```
import pandas as pd
from sklearn.feature_extraction.text import CountVectorizer
from sklearn.metrics.pairwise import cosine_similarity

# Load the dataset into a pandas dataframe
df = pd.read_excel('dataset.xlsx')

# Get all unique symptoms
symptoms = set()
for s in df['Symptoms']:
    for symptom in s.split(','):
        symptoms.add(symptom.strip())
```

```
def predict_disease_from_symptom(symptom_list):
```

```

user_symptoms = symptom_list
# Vectorize symptoms using CountVectorizer
vectorizer = CountVectorizer()
X = vectorizer.fit_transform(df['Symptoms'])
user_X = vectorizer.transform([' '.join(user_symptoms)])

# Compute cosine similarity between user symptoms and dataset symptoms
similarity_scores = cosine_similarity(X, user_X)

# Find the most similar disease(s)
max_score = similarity_scores.max()
max_indices = similarity_scores.argmax(axis=0)
diseases = set()
for i in max_indices:
    if similarity_scores[i] == max_score:
        diseases.add(df.iloc[i]['Disease'])

# Output results
if len(diseases) == 0:
    return "<b>No matching diseases found</b>",""
elif len(diseases) == 1:
    print("The most likely disease is:", list(diseases)[0])
    disease_details = getDiseaseInfo(list(diseases)[0])
    return f"<b>{list(diseases)[0]}</b><br>{disease_details}",list(diseases)[0]
else:
    return "The most likely diseases are<br><b>" + ' '.join(list(diseases))+"</b>",""

symptoms = {'itching': 0, 'skin_rash': 0, 'nodal_skin_eruptions': 0, 'continuous_sneezing': 0,
            'shivering': 0, 'chills': 0, 'joint_pain': 0, 'stomach_pain': 0, 'acidity': 0,
            'ulcers_on_tongue': 0,
            'muscle_wasting': 0, 'vomiting': 0, 'burning_micturition': 0, 'spotting_ urination': 0,
            'fatigue': 0,
            'weight_gain': 0, 'anxiety': 0, 'cold_hands_and_feets': 0, 'mood_swings': 0,
            'weight_loss': 0,
            'restlessness': 0, 'lethargy': 0, 'patches_in_throat': 0, 'irregular_sugar_level': 0, 'cough':
            0,
            'high_fever': 0, 'sunken_eyes': 0, 'breathlessness': 0, 'sweating': 0, 'dehydration': 0,
            'indigestion': 0, 'headache': 0, 'yellowish_skin': 0, 'dark_urine': 0, 'nausea': 0,
            'loss_of_appetite': 0,
            'pain_behind_the_eyes': 0, 'back_pain': 0, 'constipation': 0, 'abdominal_pain': 0,
            'diarrhoea': 0, 'mild_fever': 0,
            'yellow_urine': 0, 'yellowing_of_eyes': 0, 'acute_liver_failure': 0, 'fluid_overload': 0,
            'swelling_of_stomach': 0,
            'swelled_lymph_nodes': 0, 'malaise': 0, 'blurred_and_distorted_vision': 0, 'phlegm': 0,
            'throat_irritation': 0,
            'redness_of_eyes': 0, 'sinus_pressure': 0, 'runny_nose': 0, 'congestion': 0, 'chest_pain': 0,
            'weakness_in_limbs': 0,

```

```

        'fast_heart_rate': 0, 'pain_during_bowel_movements': 0, 'pain_in_anal_region': 0,
'bloody_stool': 0,
        'irritation_in_anus': 0, 'neck_pain': 0, 'dizziness': 0, 'cramps': 0, 'bruising': 0, 'obesity': 0,
'swollen_legs': 0,
        'swollen_blood_vessels': 0, 'puffy_face_and_eyes': 0, 'enlarged_thyroid': 0,
'brittle_nails': 0, 'swollen_extremeties': 0,
        'excessive_hunger': 0, 'extra_marital_contacts': 0, 'drying_and_tingling_lips': 0,
'slurred_speech': 0,
        'knee_pain': 0, 'hip_joint_pain': 0, 'muscle_weakness': 0, 'stiff_neck': 0,
'swelling_joints': 0, 'movement_stiffness': 0,
        'spinning_movements': 0, 'loss_of_balance': 0, 'unsteadiness': 0,
'weakness_of_one_body_side': 0, 'loss_of_smell': 0,
        'bladder_discomfort': 0, 'foul_smell_of_urine': 0, 'continuous_feel_of_urine': 0,
'passage_of_gases': 0, 'internal_itching': 0,
        'toxic_look_(typhos)': 0, 'depression': 0, 'irritability': 0, 'muscle_pain': 0,
'altered_sensorium': 0,
        'red_spots_over_body': 0, 'belly_pain': 0, 'abnormal_menstruation': 0, 'dischromic
_patches': 0, 'watering_from_eyes': 0,
        'increased_appetite': 0, 'polyuria': 0, 'family_history': 0, 'mucoid_sputum': 0,
'rusty_sputum': 0, 'lack_of_concentration': 0,
        'visual_disturbances': 0, 'receiving_blood_transfusion': 0,
'receiving_unsterile_injections': 0, 'coma': 0,
        'stomach_bleeding': 0, 'distention_of_abdomen': 0, 'history_of_alcohol_consumption':
0, 'fluid_overload.1': 0,
        'blood_in_sputum': 0, 'prominent_veins_on_calf': 0, 'palpitations': 0, 'painful_walking':
0, 'pus_filled_pimples': 0,
        'blackheads': 0, 'scurring': 0, 'skin_peeling': 0, 'silver_like_dusting': 0,
'small_dents_in_nails': 0, 'inflammatory_nails': 0,
        'blister': 0, 'red_sore_around_nose': 0, 'yellow_crust_ooze': 0}

```

Set value to 1 for corresponding symptoms

```

for s in symptom_list:
    index = predict_symptom(s, list(symptoms.keys()))
    print('User Input: ',s," Index: ",index)
    symptoms[index] = 1

```

Put all data in a test dataset

```

df_test = pd.DataFrame(columns=list(symptoms.keys()))
df_test.loc[0] = np.array(list(symptoms.values()))
print(df_test.head())
# Load pre-trained model
clf = load(str("model/random_forest.joblib"))
result = clf.predict(df_test)

```

```

disease_details = getDiseaseInfo(result[0])

```

```

# Cleanup
del df_test

```

```

return f"<b>{result[0]}</b><br>{disease_details}",result[0]

```

```
import pandas as pd
from sklearn.feature_extraction.text import CountVectorizer
from sklearn.metrics.pairwise import cosine_similarity

# Get all unique diseases
diseases = set(df['Disease'])

def get_symptoms(user_disease):
    # Vectorize diseases using CountVectorizer
    vectorizer = CountVectorizer()
    X = vectorizer.fit_transform(df['Disease'])
    user_X = vectorizer.transform([user_disease])

    # Compute cosine similarity between user disease and dataset diseases
    similarity_scores = cosine_similarity(X, user_X)

    # Find the most similar disease(s)
    max_score = similarity_scores.max()
    print(max_score)
    if max_score < 0.7:
        print("No matching diseases found")
        return False, "No matching diseases found"
    else:
        max_indices = similarity_scores.argmax(axis=0)
        symptoms = set()
        for i in max_indices:
            if similarity_scores[i] == max_score:
                symptoms.update(set(df.iloc[i]['Symptoms'].split(',')))
        # Output results

        print("The symptoms of", user_disease, "are:")
        for sym in symptoms:
            print(str(sym).capitalize())

        return True, symptoms

from duckduckgo_search import ddg

def getDiseaseInfo(keywords):
    results = ddg(keywords, region='wt-wt', safesearch='Off', time='y')
    return results[0]['body']

@app.route('/ask', methods=['GET', 'POST'])
def chat_msg():
```

```
user_message = request.args["message"].lower()
sessionId = request.args["sessionId"]

rand_num = random.randint(0,4)
response = []
if request.args["message"]=="undefined":

    response.append(msgCons.WELCOME_GREET[rand_num])
    response.append("What is your good name?")
    return jsonify({'status': 'OK', 'answer': response})
else:

    currentState = userSession.get(sessionId)

    if currentState ==-1:
        response.append("Hi "+user_message+", To predict your disease based on symptopms,
we need some information about you. Please provide accordingly.")
        userSession[sessionId] = userSession.get(sessionId) +1
        all_result['name'] = user_message

    if currentState==0:
        username = all_result['name']
        response.append(username+" , what is you age?")
        userSession[sessionId] = userSession.get(sessionId) +1

    if currentState==1:
        pattern = r'\d+'
        result = re.findall(pattern, user_message)
        if len(result)==0:
            response.append("Invalid input please provide valid age.")
        else:
            if float(result[0])<=0 or float(result[0])>=130:
                response.append("Invalid input please provide valid age.")
            else:
                all_result['age'] = float(result[0])
                username = all_result['name']
                response.append(username+" , Choose Option ?")
                response.append("1. Predict Disease")
                response.append("2. Check Disease Symtoms")
                userSession[sessionId] = userSession.get(sessionId) +1

    if currentState==2:

        if '2' in user_message.lower() or 'check' in user_message.lower():
            username = all_result['name']
            response.append(username+" , What's Disease Name?")
            userSession[sessionId] = 20
        else:

            username = all_result['name']
```

```
response.append(username+" What symptoms are you experiencing?")
response.append('<a href="/diseases" target="_blank">Symptoms List</a>')
userSession[sessionId] = userSession.get(sessionId) + 1
```

```
if currentState==3:
```

```
all_result['symptoms'].extend(user_message.split(", "))
username = all_result['name']
response.append(username+" What kind of symptoms are you currently experiencing?")
response.append("1. Check Disease")
response.append('<a href="/diseases" target="_blank">Symptoms List</a>')
userSession[sessionId] = userSession.get(sessionId) + 1
```

```
if currentState==4:
```

```
if '1' in user_message or 'disease' in user_message:
    disease,type = predict_disease_from_symptom(all_result['symptoms'])
    response.append("<b>The following disease may be causing your discomfort</b>")
    response.append(disease)
    response.append(f'<a href="https://www.google.com/search?q={type} disease hospital
near me" target="_blank">Search Near By Hospitals</a>')
    userSession[sessionId] = 10
```

```
else:
```

```
all_result['symptoms'].extend(user_message.split(", "))
username = all_result['name']
response.append(username+" Could you describe the symptoms you're suffering
from?")
response.append("1. Check Disease")
response.append('<a href="/diseases" target="_blank">Symptoms List</a>')
userSession[sessionId] = userSession.get(sessionId) + 1
```

```
if currentState==5:
```

```
if '1' in user_message or 'disease' in user_message:
    disease,type = predict_disease_from_symptom(all_result['symptoms'])
    response.append("<b>The following disease may be causing your discomfort</b>")
    response.append(disease)
    response.append(f'<a href="https://www.google.com/search?q={type} disease hospital
near me" target="_blank">Search Near By Hospitals</a>')
    userSession[sessionId] = 10
```

```
userSession[sessionId] = 10
```

```
else:
```

```
all_result['symptoms'].extend(user_message.split(", "))
username = all_result['name']
response.append(username+" What are the symptoms that you're currently dealing
```

with?")

```
response.append("1. Check Disease")
response.append('<a href="/diseases" target="_blank">Symptoms List</a>')
userSession[sessionId] = userSession.get(sessionId) +1
```

if currentState==6:

```
if '1' in user_message or 'disease' in user_message:
    disease,type = predict_disease_from_symptom(all_result['symptoms'])
    response.append("The following disease may be causing your discomfort")
    response.append(disease)
    response.append(f'<a href="https://www.google.com/search?q={type} disease hospital
near me" target="_blank">Search Near By Hospitals</a>')
    userSession[sessionId] = 10
else:
    all_result['symptoms'].extend(user_message.split(","))
    username = all_result['name']
    response.append(username+" , What symptoms have you been experiencing lately?")
    response.append("1. Check Disease")
    response.append('<a href="/diseases" target="_blank">Symptoms List</a>')
    userSession[sessionId] = userSession.get(sessionId) +1
```

if currentState==7:

```
if '1' in user_message or 'disease' in user_message:
    disease,type = predict_disease_from_symptom(all_result['symptoms'])
    response.append("<b>The following disease may be causing your discomfort</b>")
    response.append(disease)
    response.append(f'<a href="https://www.google.com/search?q={type} disease hospital
near me" target="_blank">Search Near By Hospitals</a>')
    userSession[sessionId] = 10
else:
    all_result['symptoms'].extend(user_message.split(","))
    username = all_result['name']
    response.append(username+" , What are the symptoms that you're currently dealing
with?")
```

```
response.append("1. Check Disease")
response.append('<a href="/diseases" target="_blank">Symptoms List</a>')
userSession[sessionId] = userSession.get(sessionId) +1
```

if currentState==8:

```
if '1' in user_message or 'disease' in user_message:
    disease,type = predict_disease_from_symptom(all_result['symptoms'])
    response.append("The following disease may be causing your discomfort")
    response.append(disease)
    response.append(f'<a href="https://www.google.com/search?q={type} disease hospital
near me" target="_blank">Search Near By Hospitals</a>')
    userSession[sessionId] = 10
else:
    all_result['symptoms'].extend(user_message.split(","))
```

```
username = all_result['name']
response.append(username+" , What symptoms have you been experiencing lately?")
response.append("1. Check Disease")
response.append('<a href="/diseases" target="_blank">Symptoms List</a>')
userSession[sessionId] = userSession.get(sessionId) +1

if currentState==10:
    response.append('<a href="/user" target="_blank">Predict Again</a>')

if currentState==20:

    result,data = get_symtoms(user_message)
    if result:
        response.append(f"The symptoms of {user_message} are")
        for sym in data:
            response.append(sym.capitalize())

    else:response.append(data)

    userSession[sessionId] = 2
    response.append("")
    response.append("Choose Option ?")
    response.append("1. Predict Disease")
    response.append("2. Check Disease Symtoms")

return jsonify({'status': 'OK', 'answer': response})

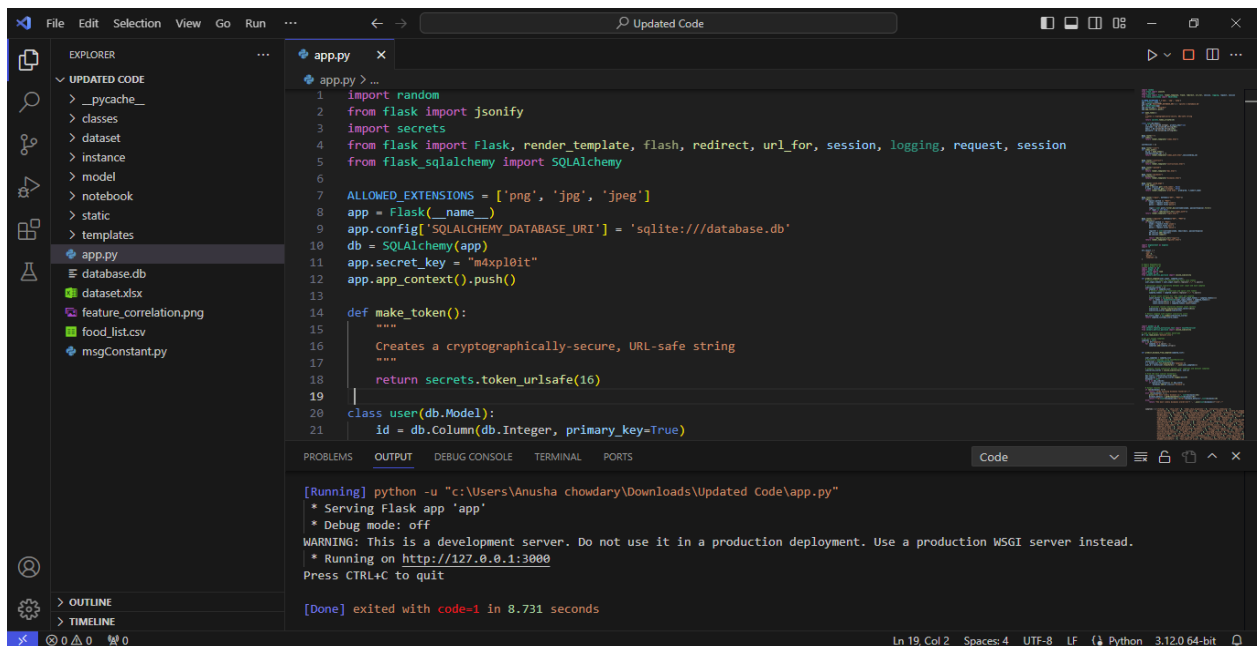
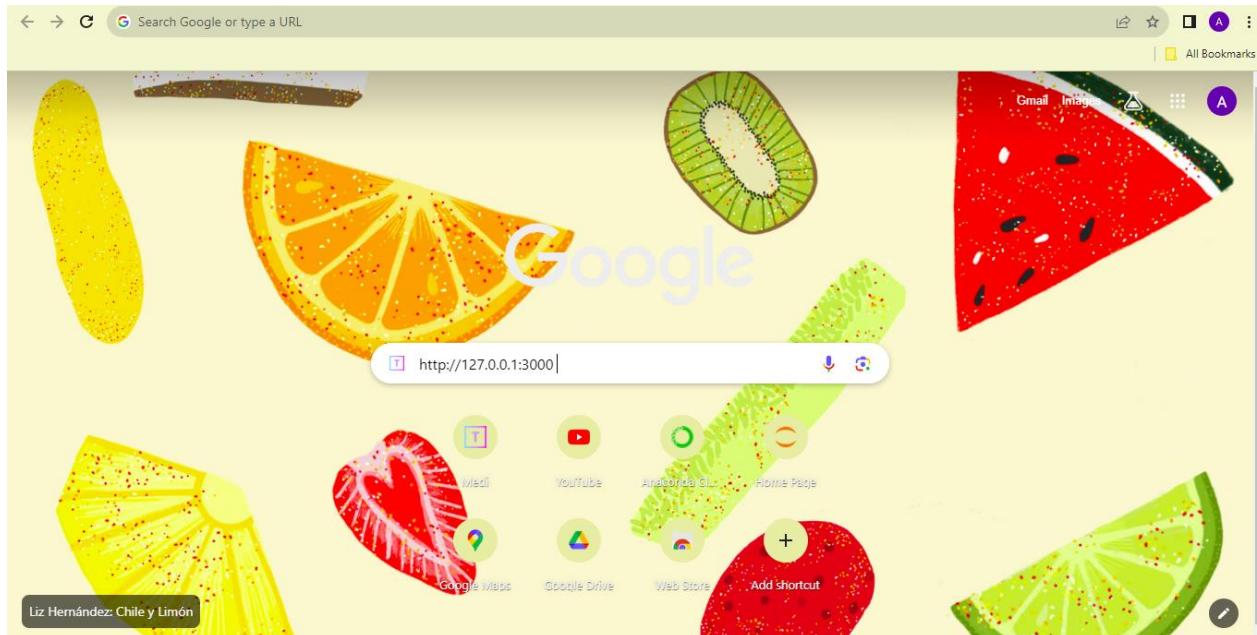
if __name__ == "__main__":

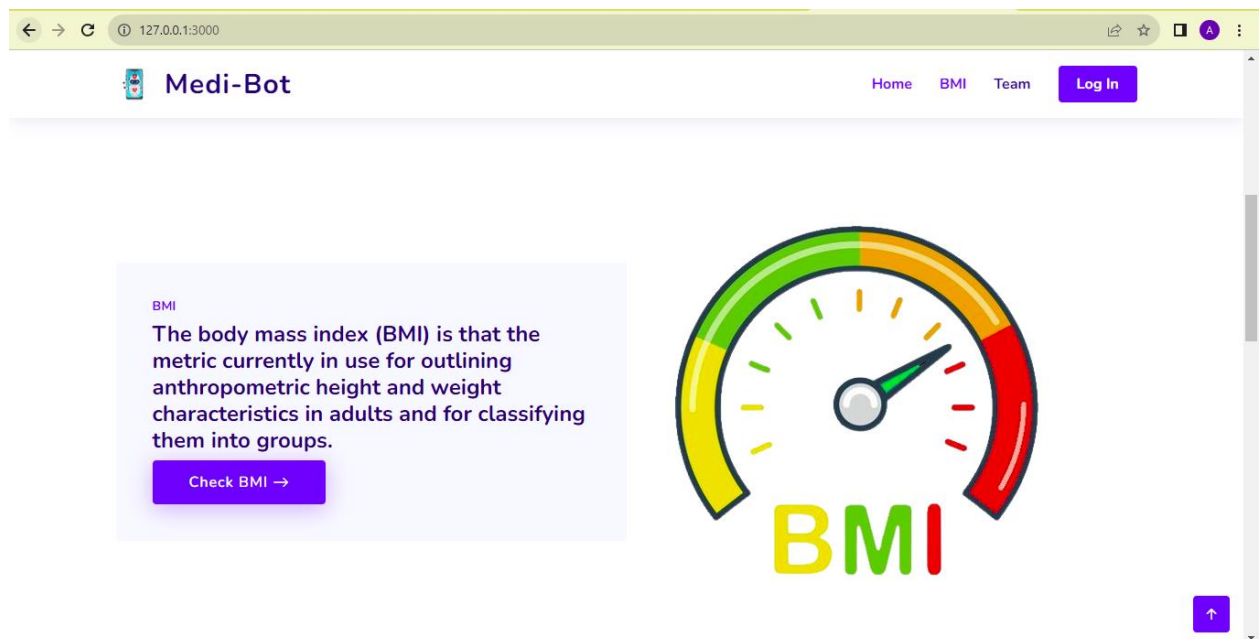
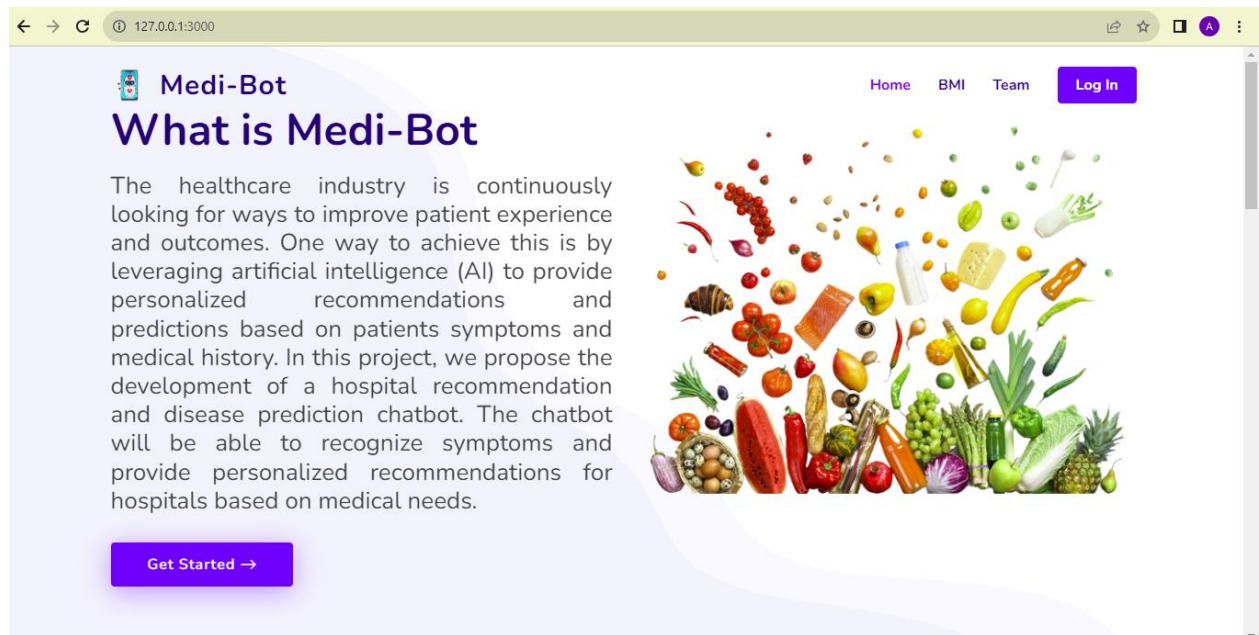
    db.create_all()
    app.run(debug=False, port=3000)
```

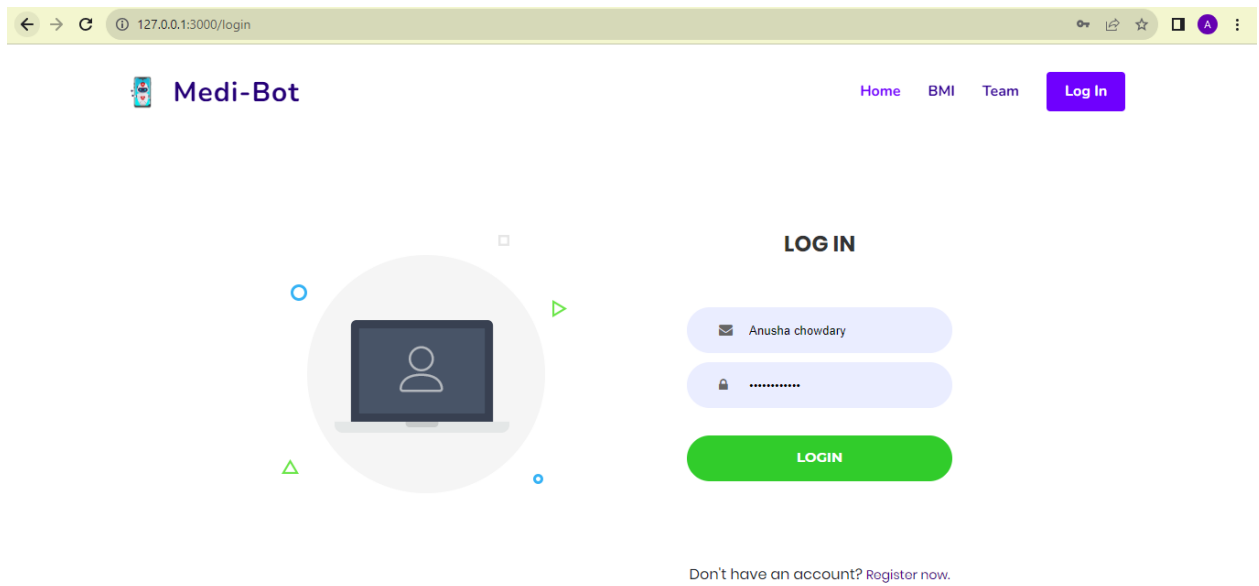
CHAPTER 9

SNAPSHOTS

9. SNAPSHOTS







The screenshot shows the login page of the Medi-Bot application. The browser's address bar displays "127.0.0.1:3000/login". The page features a navigation bar with links for "Home", "BMI", "Team", and a "Log In" button. On the left, there is a graphic of a laptop with a user icon, surrounded by four colored circles (blue, green, yellow, and red). The main heading is "LOG IN". Below this, there are two input fields: the first is labeled "Anusha chowdary" and the second is masked with "*****". A green "LOGIN" button is positioned below the password field. At the bottom, a link reads "Don't have an account? Register now."

Medi-Bot

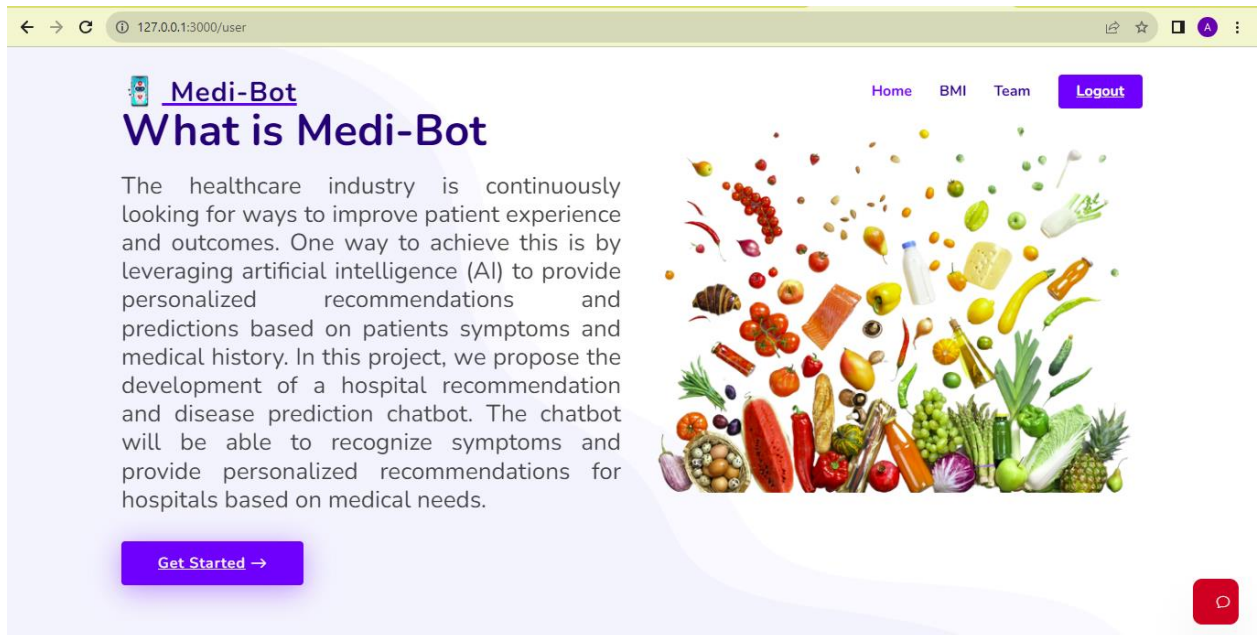
Home BMI Team Log In

LOG IN

Anusha chowdary

LOGIN

Don't have an account? Register now.



The screenshot shows the "What is Medi-Bot" page. The browser's address bar displays "127.0.0.1:3000/user". The navigation bar includes "Home", "BMI", "Team", and a "Logout" button. The heading "What is Medi-Bot" is prominently displayed. The text describes the project's goal: to improve patient experience and outcomes using AI for personalized recommendations and predictions based on symptoms and medical history. A "Get Started →" button is located at the bottom left. On the right, there is a large, vibrant image of various fruits and vegetables. A red chatbot icon is visible in the bottom right corner.

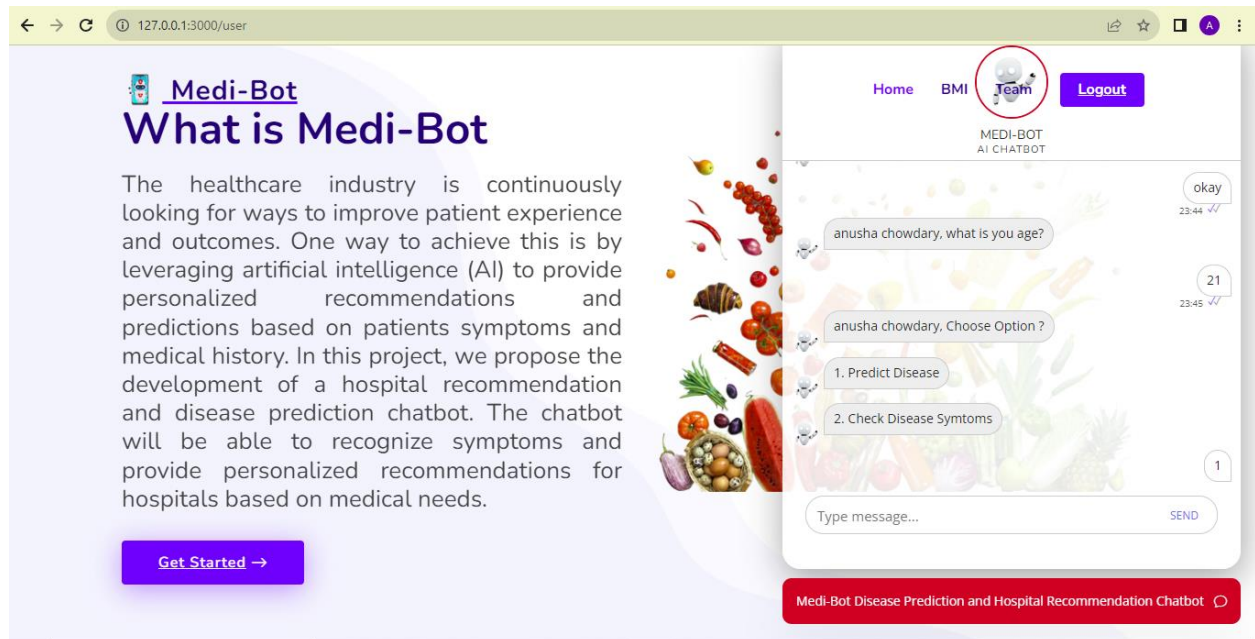
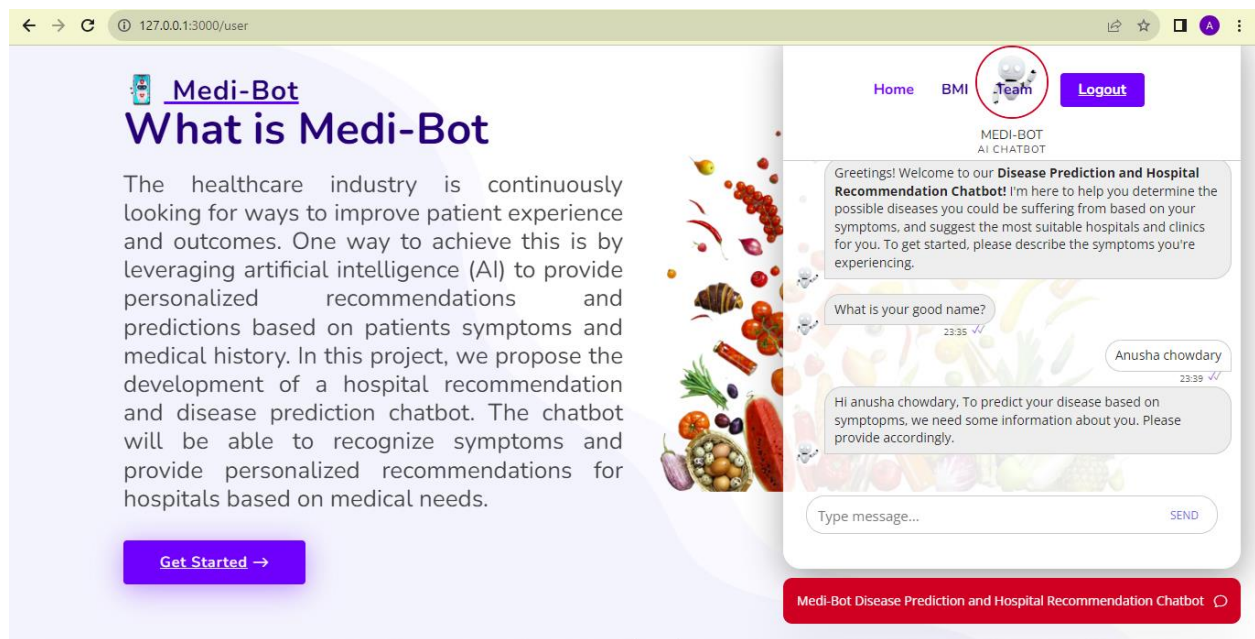
Medi-Bot

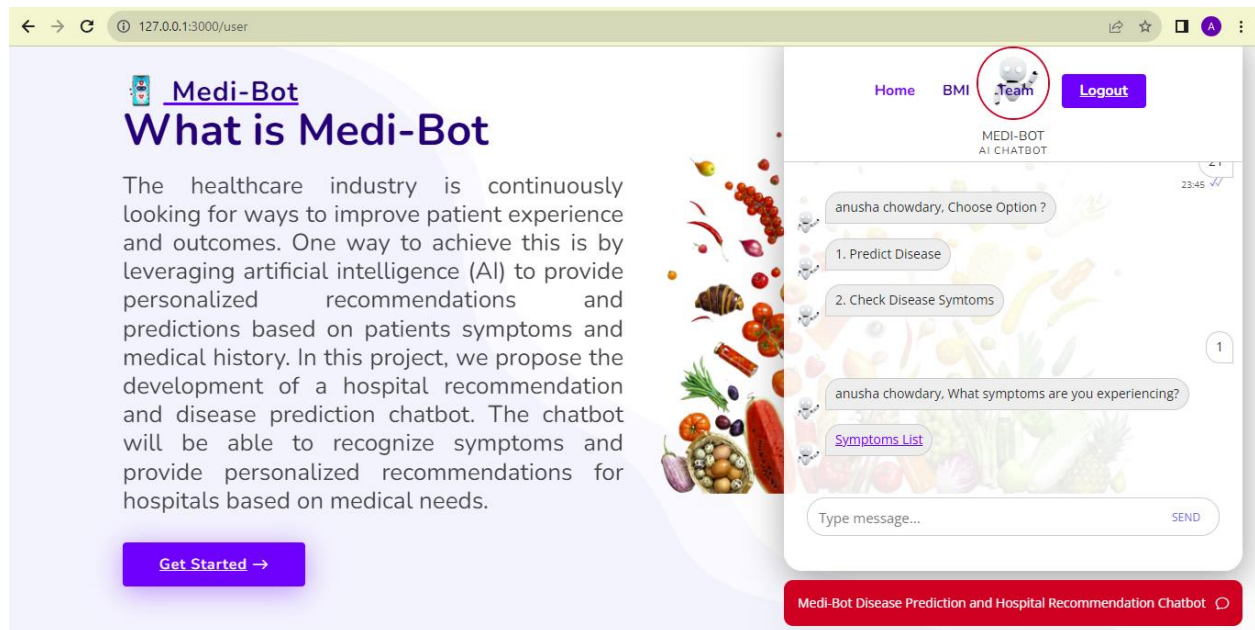
Home BMI Team Logout

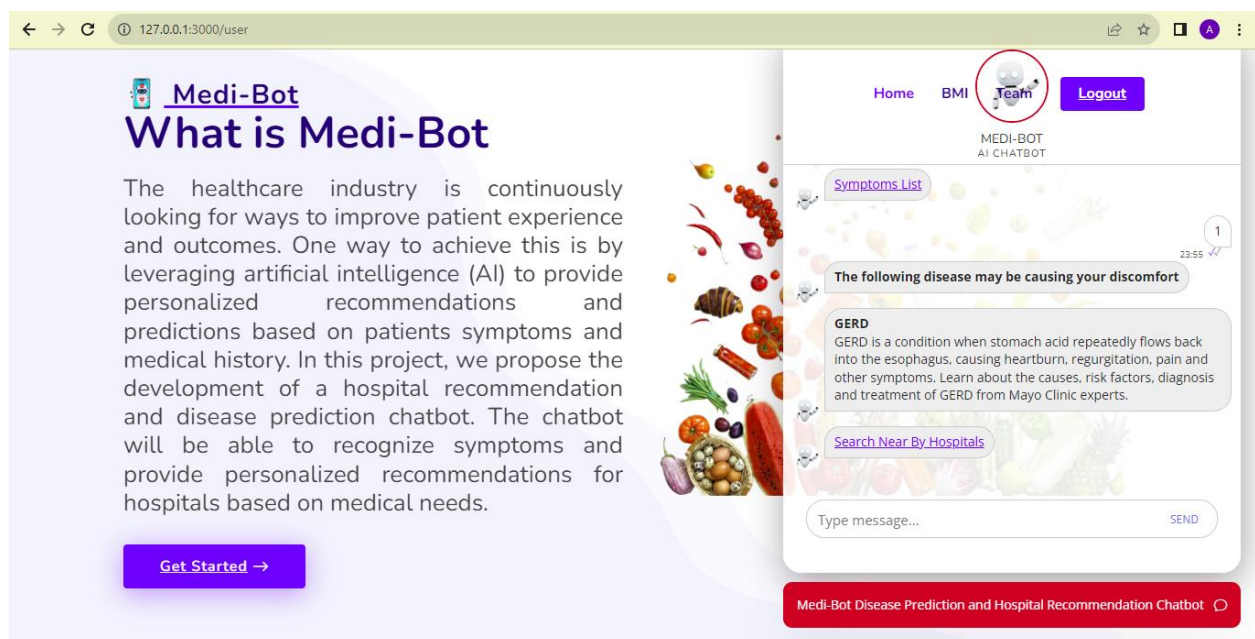
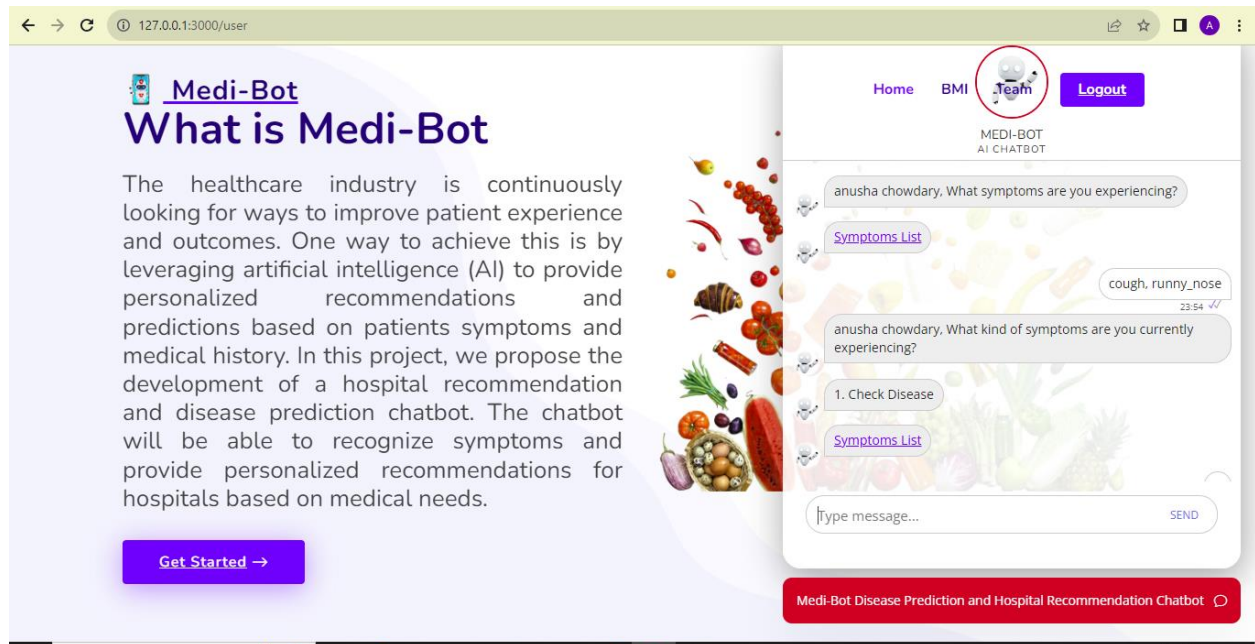
What is Medi-Bot

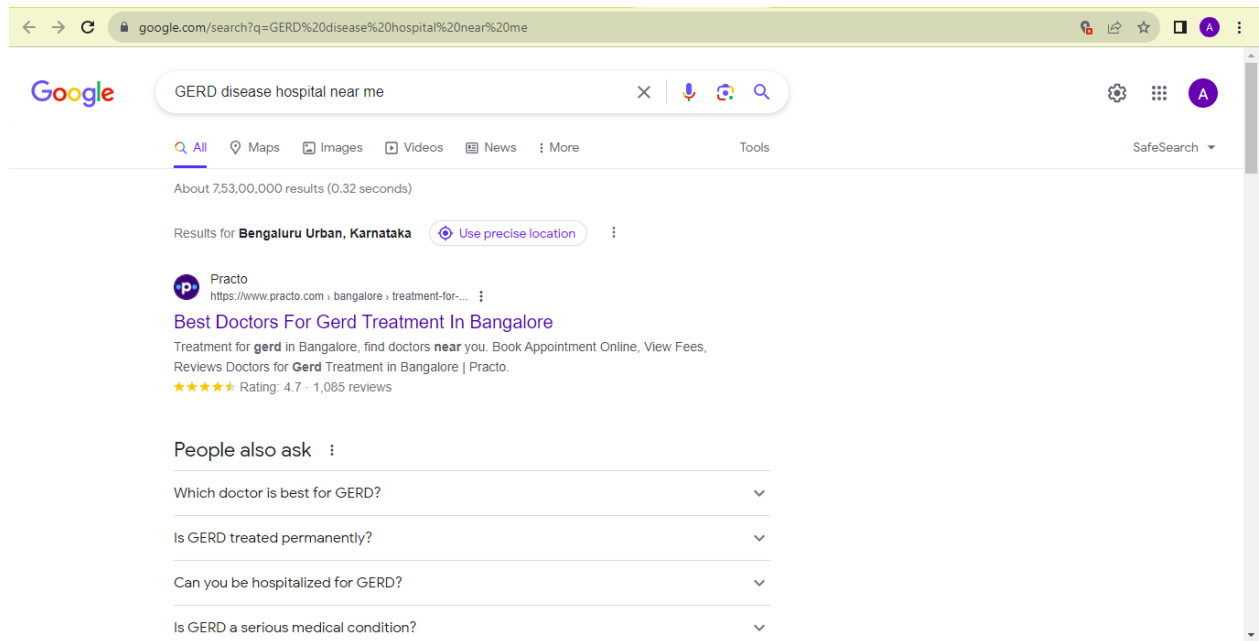
The healthcare industry is continuously looking for ways to improve patient experience and outcomes. One way to achieve this is by leveraging artificial intelligence (AI) to provide personalized recommendations and predictions based on patients symptoms and medical history. In this project, we propose the development of a hospital recommendation and disease prediction chatbot. The chatbot will be able to recognize symptoms and provide personalized recommendations for hospitals based on medical needs.

Get Started →









CHAPTER 10

CONCLUSION

10. CONCLUSION

The package was designed in such a way that future modifications can be done easily. The following conclusions can be deduced from the development of the project:

- ❖ Automation of the entire system improves the efficiency
- ❖ It provides a friendly graphical user interface which proves to be better when compared to the existing system.
- ❖ It gives appropriate access to the authorized users depending on their permissions.
- ❖ It effectively overcomes the delay in communications.
- ❖ Updating of information becomes so easier
- ❖ System security, data security and reliability are the striking features.
- ❖ The System has adequate scope for modification in future if it is necessary.

REFERENCE

1. <https://www.geeksforgeeks.org/introduction-machine-learning/>
2. https://www.researchgate.net/publication/267575064_Using_dialogue_corpora_to_train_a_chatbot
3. https://www.researchgate.net/publication/343057048_Conversational_Chatbot_with_AI_ML

